Players and Performance: Opportunities for Social Interaction with Augmented Tabletop Games at Centres for Children with Autism

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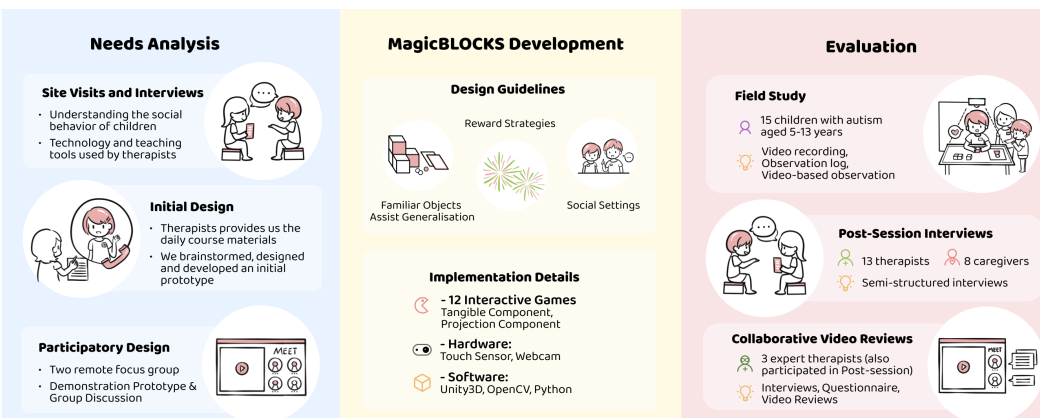

Fig. 1. Overview of Need Analysis, MagicBLOCKS Development and Evaluation.

This research aimed to investigate how children with autism interacted with rich audio and visual augmented reality (AR) tabletop games. Based on in-depth needs analysis facilitated through autism centers in China, we designed and developed MagicBLOCKS, a series of tabletop AR interactive games for children with autism. We conducted a four-week field study with 15 male children. We found that the interactive dynamics in games were rewarding and played critical roles in motivation and sustained interest. In addition, based on post-hoc interviews and video analysis with expert therapists, we found that MagicBLOCKS provided opportunities for children with autism to engage with each other through player performances and audience interactions with episodes of cooperation and territoriality. We discuss the limitations and the insights offered by this research.

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CCS Concepts: • **Human-centered computing → User interface toolkits**; **Accessibility systems and tools**.

Additional Key Words and Phrases: Children with autism, Tabletop Interaction, Augmented Reality(AR), Audience Socializing, Assistive Technology, Accessibility

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1 INTRODUCTION

Social skills enable us to successfully navigate rules and norms that govern our interactions with others and the environment. In general, people can learn social skills in similar ways to learning language skills [28, 40]. Children with autism develop differently than neurotypical peers. Those with autism engage in different social behaviors, such as avoiding looking at the eye-region when interacting with others, which is related to a very strong sensory response and the cause is as yet undetermined [44]. Peer feedback significantly affects social decision-making in autistic individuals, particularly during adolescence [48]. A relaxed environment might enable children with autism to collaborate with peers, communicate, and learn from one another [12]. Children with autism display extraordinary tenacity and persistence in specific settings [15, 29]. Rewarding positive behaviors in autistic children is a common practice [3]. Reinforcement through rewards encourages children with autism to repeat desired behaviors [10]. This includes positive reinforcers such as verbal praise and access to toys they like.

Several studies have shown that interactive technology can enhance social interaction for children with autism [22, 45, 50–52]. Recently, there has been a growing interest in developing augmented reality interactions to support children with autism in learning skills [5, 21, 30, 54]. To fully realize the potential of this type of workspace, we must first understand the real needs of various stakeholders. We also need to observe the behaviors of children with autism when playing interactive games. This paper presents our exploratory, qualitative study with children with autism aimed to understand their interactions around and through augmented tabletop games.

We approached the design of interactive tabletop gameplay for children with autism using a general user-centred design methodology. We began with research to understand the context, which included in-depth interviews with therapists and caregivers of children with autism, as well as site visits and observations of how these children participate in daily education. First, we conducted a formative study that included face-to-face interviews with therapists and caregivers of children with autism as well as observations of these children as they engaged in their daily training. These findings have been synthesized to create our design guidelines. Based on these principles, we developed MagicBLOCKS, an interactive tabletop n augmented reality system. It includes 12 interactive games. The system uses tangible components (such as building blocks, cards, colored balls, and toothbrushes) for input and digital projections (such as colors, sounds, and animations) for output. We conducted a field study to observe interactions afforded by the system. An overview of our work is shown in Figure 1.

Our findings indicate that the MagicBLOCKS system provides children with autism with motivation and opportunities to engage with each other. The rich audio and visual qualities of the augmented tabletop game were rewarding and played a key role in motivation and sustained interest. Themes around social dynamics emerged from our observations. Children with autism engaged through player performances and audience interactions with cooperation and territoriality. We discuss ways to improve tabletop interactive games to increase opportunities for social interaction for children with autism, as well as how these findings inspire design insights within the ISS community. In summary, our contributions include:

- Insights from a formative study with therapists, caregivers, and children with autism that lead to design guidelines and technical implementation of MagicBLOCKS system.
- Results from a field study with MagicBLOCKS demonstrating how tabletop augmented reality interactive games can be rewarding and can provide children with autism opportunities for social interaction.

2 RELATED WORK

2.1 Assistive Technology for Children with Autism

Technological innovation can be used to assist and support the interaction of individuals with autism [4]. Assistive technology can help children with autism in learning essential skills [1]. Among various approaches in assistive technology, physical interaction technologies, such as tangible interfaces, are widely used to support children with autism. Frauenberger et el. use a variety of participatory design methods to create a series of tangible intelligent hardware with children with austism [13, 14, 41]. Sitdhisanguan et al. studied the benefits of Windows, Icons, Menus and Pointing devices (WIMPs) and tangible user interfaces (TUIs) for children with autism and their results showed that TUI-based interaction achieved higher skill improvement compared with WIMP-based interaction [38]. There is growing evidence that augmented reality technology has the potential to assist autistic children. For example, 'Let's Cook' is an augmented reality game designed to develop cooking skills for children with cognitive needs [30]. 'FUTUREGYM' is a gymnasium designed for children with autism that features interactive floor projections [43]. The 'Collaborative Puzzle Game' is a tabletop interactive cardboard jigsaw activity that promotes collaboration among children with autism through collaborative movement, i.e., in order to move a puzzle piece, it must be touched and dragged simultaneously by two players [2]. Our research is similar to the 'Collaborative Puzzle Game' in that we are pursuing tabletop interaction, and we differ as we will also include AR and we will enable flexible solo or collaborative play.

The studies 'Let's Cook' and 'FUTUREGYM' found that augmented reality can benefit autistic children's learning, especially when games have incorporated interactive tangible components. These prior works inspired our system design, which motivated us to include augmented reality and tangible components. Our research aims to investigate how rich auditory and visual interactions could support motivation and sustained interest in children with autism. We were also interested in how games may facilitate social opportunities.

2.2 Technologies to Support Social Interaction for Children with Autism

Numerous previous design studies have been conducted to to support the social interactions of children with autism. Robots have been found to be helpful [27, 32, 35, 42]. Focus on design to support interaction has also helped. For instance, Hayes' team designed the concept of immediacy of interaction to support classroom social interaction in the daily life of children with autism in public schools [19, 45]. Li et al. designed a mobile game platform that uses social and non-social stimuli to assess executive function (EF) abilities of children with autism and they found that it was able to predict children's IQ [26]. 'MOSOCO' is a smartphone assistive technology that combines virtual reality and visual aids from a validated program, 'the Social Compass', to assist children with autism in practising social skills in real-world settings [8, 9]. 'SuperpowerGlass' is a wearable device with automatic facial expression recognition running on Google Glass, which can provide real-time social prompts for children with autism during home treatment [22, 50, 51]. 'MaskMe' is a wearable smart mask that supports children with autism in maintaining eye contact during social activities by creating collaborative games [53]. The music user interface Reactable was used to probe the social interactions of children with autism and the findings indicated that participants had increased social turn-taking, even in non-verbal communication [49]. Wu et al. developed a set of interactive playgrounds comprised of interactive softballs to assist children with autism with social activities [52]. These projects looked at how children with autism engage with similar peers and neurotypical peers. We also want to examine how interactive tabletop games may be able to support children with autism in engaging socially.

2.3 Autistic People and Audiences

The audience effect is defined as occurring when a person is observed by others or believes that they are being observed by others [18, 47]. A Japanese research team attempted to train participants with autism in public speech using a simple virtual audience. By utilizing a virtual audience, public speaking pressure was reduced and participants' self-confidence was improved [25]. Fletcher-Watson described relaxed performances, created for Autistic audiences, where audience members are allowed to move around, and there are fewer bright lights and quieter sounds and found these were found to be comfortable to audience members and conducive to meeting others in the audience [11]. Children with autism are influenced by their peers around them. Children with autism engaged in less repetitive behaviors and more verbal operants (verbal social interaction behaviors) in the presence of neurotypical peers compared to special education peers [37]. We are also interested to observe peer interactions, in particular those facilitated by augmented reality tabletop interactions.

In summary, there is a large body of research focusing on technology tools to support children with autism in social interaction, with some looking at the role of an audience member. To our knowledge, to date no studies have been undertaken on the observation of social interactions among children with autism facilitated by augmented reality tabletop space with tangible components. We adopted a user-centred design method to elicit requirements and develop an augmented tabletop game and evaluated it with a qualitative analysis to shed light on these topics.

3 NEEDS ANALYSIS AND SYSTEM DESIGN

3.1 Design Methods

3.1.1 Site Visits and Interviews. We conducted site visits and interviewed therapists and parents from three autism centers in different cities. These organisations provide services, support and advocacy for children with autism. These site visits aimed to better understand the context and the challenges of social settings for children with autism. The therapists we interviewed had extensive work experience in social skills training. The parents we interviewed had long-term experience in taking care of children with autism. We visited these places two to three times, observed the interactions of children with autism and took field notes and photographs from 2019 to 2020. Specifically, we focused on understanding the behavior of children with autism in peer social activities in the public space. We paid particular attention to understand the technology and teaching tools that therapists use in the training space. In addition, we conducted video reviews with three special education teachers, five therapists and five caregivers to understand their needs and perspective during the COVID.

3.1.2 Initial Design. Based on the findings from the site visits and interviews, our team worked on an initial prototype. The core design and programming team included one graduate student and three undergraduate students. As input to our design work, we asked therapists to send us course materials that they used daily, such as daily checklists for learning, course planning for gameplay, lesson plans for cognitive practices, and strategies for team activities. We also consulted additional autism learning strategies and tools online. Our core team brainstormed, designed and developed the games' initial interactions.

3.1.3 Participatory Design. Building on the relationships during our initial site visits, we conducted a semi-remote participatory design in the spring of 2020. Our therapist participants suggested that the autistic children might have difficulty participating in remote communication so we did not involve them in this phase (though doing research without direct stakeholders involved is not recommended [41] and we discuss this more in our limitations). In the two remote focus group meetings with four expert therapists, we showed our MagicBLOCKS prototype together with a video that showed how children could interact with the system. After watching the video, we conducted a group discussion.

3.2 Design Guidelines

Insights from the focus group discussion, previous site visits, and interviews informed a set of core design considerations, leading to the MagicBLOCKS system's final design. Here we describe three design guidelines for MagicBLOCKS.

3.2.1 Familiar Objects Assist Generalisation. Tangible objects, such as building blocks, cards, colored balls, and visual cards are teaching tools for the daily training of children with autism. Therapists use them due to their innate appeal and proven effectiveness in supporting hand-eye coordination, which is transferable to other knowledge. The therapists recommended that we design based on items familiar to children with autism. Therefore, in MagicBLOCKS system, we included familiar tangible objects to support various interactions.

3.2.2 Reward Strategies. Rewarding positive behaviors in children with autism is a common practice used in reinforcement in Applied Behavioral Analysis (ABA) [3]. Our participant therapists indicated that common positive reinforcers they used were include desired toys, favorite foods, tokens, and verbal encouragement. The therapists were very interested in finding reinforcers that were motivating. As such, MagicBLOCKS should utilize sounds, motion and special effects (that would sustain children's interest), and combine the effects with physical objects.

3.2.3 Social Settings. Therapists explained that their typical sessions with children consist of one-on-one or multi-person social games, but that they rarely address social behaviors between two children because it was generally challenging to facilitate close social relationships with peers. Training children to deal with close social relationships with peers is one of the critical training goals of therapists at autism centres. Thus we focus on a tabletop space so that multiple children can be present simultaneously and interact, and the therapists can observe peer behavior. Therapists anticipated that they would be able to learn from their observations of children, to create subsequent social skills training programs that would be especially useful and appropriate given the observed behavior.

Taking the above design considerations into account, we developed an interactive tabletop system MagicBLOCKS to support children with autism to facilitate social interaction.

3.3 MagicBLOCKS System

MagicBLOCKS is an augmented interactive tabletop system able to support various interaction modes, such as clicking, pressing, dragging and air gestures. The system includes 12 interactive games with a broad range of topics and effects as shown in Table 1. In this table, 'Monster', 'Cannon', and 'Paint' games can be used as a warm-up experience for children. These three games can be played by clicking virtual icons (without tangible props). The 'Teeth', 'Piano', and 'Ball' games have increasing complexity. The 'Teeth' game allows users to use a physical toothbrush to swipe and click on projected toothbrush motions. This is intended to emphasize the importance of brushing one's teeth. The 'Piano' game is designed to teach numbers and musical notation. The 'Ball' game is designed to enable practice of color matching and sorting. The 'Occupation', 'Puzzle', and 'Ocean' games require players to place and arrange tangible blocks. This is intended to support practice of concentration and hand dexterity. The games 'Maze', 'Card', and 'Cash' are designed to help develop logic skills, item matching, and banknote recognition. Our games were updated over the field trail sessions. In the first session we offered 9 games, the second session we offered 10 games, the third and last sessions we offered 12 games. Based on the feedback from the therapists and children, we updated interface details, like enlarging pictures, adding more animations, clearer sounds and more rotation motions.

Tangible objects typically used as daily teaching tools are included as system input; these include colored balls, building blocks, cards, a toothbrush and a banknote. The output is a combination of digital animation and sound effects that serve as rewards to sustain children's attention, as was recommended by our therapist participants. They advised that children with autism enjoy rotating and repetitive digital animations thus we include these in our games. For example, in the occupation game, the tabletop projection is divided into two areas: the left side shows an animation of a life scene (for example, a little boy is coughing), and the right side shows an identification box. Four blocks with patterns on six sides are provided to the player to create the pattern of the relevant profession (in this case, a doctor). If the completed blocks with the correct pattern are placed in the recognition box, the system will play an animation of the successful completion of the game (e.g., the little boy being cured, waving and saying thank you). The simplest games 'Monster', 'Cannon', and 'Paint' were designed as warm-up games that do not require any training. Before playing our more complex games (such as 'Occupation', 'Card', 'Puzzle', 'Ball') we provided a brief training session with clear instructions. To investigate social interactions, we placed the augmented tabletop in an area where we could accommodate several peers around the tabletop. We set up the games so that we were able to observe the main player as well as interactions with peers.

3.4 Implementation Details

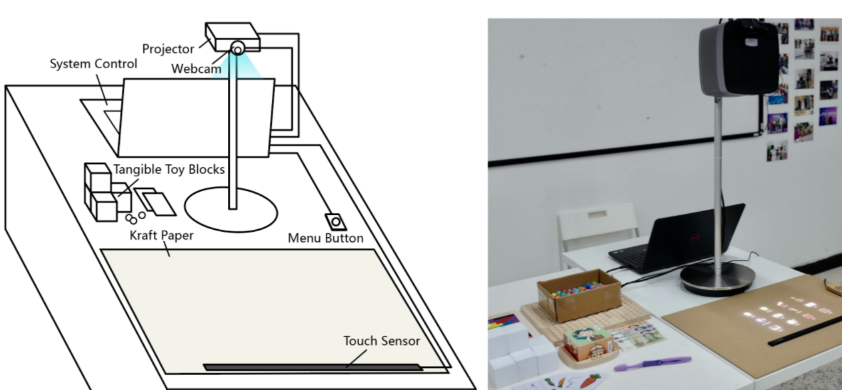

Fig. 2. The architecture diagram (left) and classroom (right) setup of the MagicBLOCKS system.

Figure 2 illustrates the system setup of MagicBLOCKS. The system occupies an area of $3m \times 4m$ and consists of four hardware components: an Airbar sensor [1] for touch sensing, a Logitech webcam [2]

[1]https://air.bar/
[2]https://www.logitech.com/en-sg/webcams

Table 1. Overview of the 12 games available in MagicBLOCKS

Name	Tangible Component	Projection Component	Description
Occupation	Toy blocks	Animation, sound	Organize four blocks to enable players to visualize an archetype of six professions (doctor, firefighter, teacher, police, courier, barber).
Card	Physical card	Animation, sound	The game is primarily used to assist children in comprehending the logical relationship between items. The players select the corresponding cards based on the animation prompts in the projection area.
Puzzle	Toy blocks	Motion, sound	Players place colorful blocks to form the pattern suggested by the dashed projection. After completing the puzzle, the firework animation is the reward.
Cash	Banknotes	Motion	Recognize and match physical banknotes that match the projection area to teach children to recognize RMB banknotes.
Ocean	Toy blocks	Animation	Children can explore and learn about marine animals by arranging white physical building blocks to project marine animals of various sizes.
Ball	Colored Ball	Motion, sound	Players can create a rotating effect on the Ferris wheel by placing a single colored ball in the groove of the wooden board.
Teeth	Toothbrush	Animation, sound	The player uses a physical toothbrush to clean the projected teeth, educating children about the importance of dental hygiene.
Monster	-	Motion, sound	By tapping the cannonballs to kill monsters coming from four directions (up, down, left and right), this game allows multiplayer collaboration.
Cannon	-	Motion, sound	Launch cannonballs to knock down colorful blocks by clicking on the tabletop, aiming for specific targets based on color, number, or size.
Piano	-	Motion, sound	Players can play the "Little Star" piano music and follow the numbered notation's prompts by clicking the keys in sequence.
Paint	-	Motion	Players can paint and create with their fingers, and the lines are colored gradients.
Maze	-	Motion	Players can drag the maze's edge to alter the path the ball takes to reach the destination.

for image recognition, a high-definition projector XGIMI H1 [3] for displaying the Unity-3D user interface, and a laptop for system control. Several toy blocks, physical cards, and tips cards are initially placed on the table edge. The core software component of MagicBLOCKS is an image recognition pipeline built on top of the OpenCV library. Specifically, the pipeline first converts the captured webcam image from RGB (red, green, blue) color space into HSV (hue, saturation, value) color space. The HSV space abstracts color by separating it from saturation and pseudo-illumination, making our system more robust towards external lighting changes. Afterward, our pipeline applies adaptive color thresholding on the HUE channel to segment the webcam image into different

[3] https://www.projectorcentral.com/XGIMI-H1.htm

components. For each component, a Sobel filter is applied for edge detection and block recognition. To facilitate the localisation of a block, we divide the desk area into a 10×10 grid and each cell is color-coded.

4 EVALUATION OF MAGICBLOCKS

We conducted a user study to observe how children with autism interacted with the MagicBLOCKS system. We investigate the following research questions:

- Can the rich auditory and visual effects of augmented reality tabletop interaction be rewarding for children with autism? In particular, will they support motivation and sustained interest?
- Can MagicBLOCKS interactions enable social interaction between children with autism? What kinds of social interactions?

The study consisted of four observation sessions where participating children interacted with MagicBLOCKS system. This was followed by post-session interviews with therapists and caregivers. Session recordings were first analysed by the research team. Then, we conducted expert video reviews with senior therapists to further understand our observations. For motivation, we were looking to observe children's behaviors that expressed desire to use the system. For sustained interest, we focus on behaviors showing the desire for sustained use, such as completing additional trials and expressing the desire to play or to have extra turns for the system. For social interactions, we were interested in all interactions between all participants throughout all the sessions.

The process of our study (see Figure 3) consists of four steps. This study was approved by a local university Institutional Review Board (IRB), and all identifiable data we collected were stored and protected in an IRB-compliant system.

4.1 Participants

We recruited 15 volunteer participants (age N between 5 to 13, see Table 2). They were identified as children with autism by the local hospital and autism centre based on their intelligence quotient, social communication, verbal skills, and physical behavior [6, 20, 34]. We welcomed participants to join us with the permission of their parents and the autism centre staff. Participation was completely optional and children were allowed to withdraw at anytime without giving a reason. Before each study, we printed the children's full name and pasted them on their clothing as recommended by the therapists. During the study, we addressed participants by their full names rather than nicknames, as this was recommended to us so that the children feel respected.

4.2 Procedure

We set up the video recorder, the MagicBLOCKS system, and tangible toy blocks before the participants entered the classroom. There were 8 researchers to help facilitate the sessions. Three

Fig. 3. Process of evaluation: 1) Field observations from the session (left:practice, right: gameplay). 2) Post-session debrief interviews with 13 therapists. 3) Video analysis of gameplay. 4) Expert review with 3 senior therapists.

Table 2. Details of the children with autism who participated in our study

ID	Support Needs	No. of Sessions	Age(years)	Gender	School
P1	Moderate needs	4	12	Male	Public elementary school
P2	Moderate needs	4	8	Male	Public elementary school
P3	Moderate needs	2	12	Male	Public elementary school
P4	Moderate needs	4	8	Male	Private elementary school
P5	Low needs	2	8	Male	Public elementary school
P6	Moderate needs	2	8	Male	Public elementary school
P7	High needs	4	9	Male	Public elementary school
P8	Moderate needs	3	8	Male	Public elementary school
P9	Low needs	3	12	Male	Public elementary school
P10	High needs	3	8	Male	Special education school
P11	Low needs	3	12	Male	Public elementary school
P12	High needs	1	9	Male	Special education school
P13	Moderate needs	1	5	Male	Private kindergarten
P14	Low needs	1	7	Male	Public elementary school
P15	Low needs	1	13	Male	Public elementary school

researchers were available to guide the participants in using the system. Two researchers led to the system demonstration and instructed participants to play games. The remaining three researchers provided timely troubleshooting assistance. Child participants and caregivers (who were not actively playing the MagicBLOCKS games) could gather around the tabletop to watch, as they desired. In other words, parent and children participants were able to come and go from the setup as they spontaneously chose. Within the analysis phase, we began to refer to the participant in the playing role as the player and we referred to the participants watching as the audience members. In addition, a therapist stood in the corner and offered expert knowledge and hands-on assistance, which we explain more about in the next paragraph.

Before the sessions, each child participant received a plastic container with their name on it for storing game-play bonus gifts. These gifts included chocolates, candies, stickers, night lights, fans and small toys. These items were selected based on suggestions from therapists. With the therapists, we agreed on the following guidelines for giving gifts: when a child completed a game, they could get a candy or a small toy (whichever they preferred), and, when each session was complete we would give a sticker.

The therapists were always present during sessions and the therapists were in charge of reacting and coordinating an immediate response to any child showing distress. In these cases, therapists would sometimes give a gift to the child showing signs of distress or would instruct the researcher to give a gift to the child showing signs of distress. Child participants were given toy blocks and physical cards to practice with, to guide the researchers on whether the tangible object was appropriate for their level of dexterity. After this, the participant was led by another researcher to the table to play MagicBLOCKS. Each participant played for approximately 25 minutes. We would stop the game immediately if the participant was not interested in the system or engaged in distressed behaviors. We extended the session time by 5 to 10 minutes if the participant was engrossed in playing. To complete sessions for all 15 participants, we conducted a total of four site visits between November 2020 and January 2021. Each visit lasted approximately 3.5 hours. To facilitate more children participating in the game experience, we set up two devices during the fourth session.

4.3 Post-Session Interviews

The main sessions were followed by post-session interviews with the caregivers and therapists, to best interpret children's experiences and behaviors. We invited 13 therapists with 2 to 10 years of work experience and 8 caregivers of children with autism (see Table 3) to these post-session debrief interviews, which were semi-structured. (Two of the therapists: Th1 and Th2, also participated in the Needs Analysis.) The interview outline mainly focused on the children's behavior in the game. We asked about any differences between the children's behaviors when using MagicBLOCKS and their typical daily behaviors. We asked caregivers to comment on children's use of smart devices in their daily lives, in particular learning tools and gaming devices. We recorded audio and notes of the interviews.

Table 3. Details of of caregivers and therapists who participated in post-session interviews

ID	Gender	Role	Age (years)	No. of interviews	Total length (mins)
C1	Male	Father	40	1	26
C2	Female	Mother	38	2	51
C3	Female	Mother	43	2	42
C4	Female	Mother	37	1	22
C5	Male	Father	42	1	25
C6	Female	Grandmother	72	1	22
C7	Female	Mother	43	2	42
C8	Female	Mother	37	1	21
Th1	Male	Therapist	37	4	132
Th2	Female	Therapist	38	3	71
Th3	Female	Therapist	29	2	45
Th4	Female	Therapist	33	2	38
Th5	Female	Therapist	30	1	22
Th6	Female	Therapist	37	1	25
Th7	Female	Therapist	32	1	21
Th8	Female	Therapist	28	1	24
Th9	Female	Therapist	37	1	15
Th10	Female	Therapist	31	1	12
Th11	Female	Therapist	32	1	25
Th12	Female	Therapist	33	1	18
Th13	Female	Therapist	39	1	15

4.4 Video Review with Expert Therapists

Three therapists who participated in the post-session interviews also participated in our expert video reviews (see Table 4). These therapists also participated in our post-session interviews. We separated video clips of each child and invited the therapist to watch them one by one, so that therapists could comment on children's behaviors compared to their usual baseline behavior. We also inquired about their opinions on the social interaction facilitated by augmented reality tabletop interaction.

The video reviews with the expert therapists consisted of four sections. The first section elicited background information on the participants' relevant experience. The second section elicited therapists' comments and elaborations on the preliminary results of our video analysis. The third

Table 4. Additional details of three expert therapists who participated in the video review. (These are the same three therapists as listed in the previous table.)

ID	Specialities	Working years	Role
Th1	Play, Movement, Sensory	12	Co-founder of the Autism Center
Th2	Applied Behavior Analysis, Sensory	13	Co-founder of the Autism Center
Th3	Applied Behavior Analysis, Cognitive	5	Leader of the Autism Center

section focused on the interviewee's attitudes regarding children's behavior in the study, one by one. Open-ended questions were asked in the fourth section to inquire about their opinions on social interactions facilitated by tabletop interactive games. Finally, the survey respondents were also asked to provide general comments on MagicBLOCKS.

4.5 Data Analysis

Our data sources included field observations, post-session interviews with therapists and caregivers, and expert video review with therapists. There were 14 hours of video (123.93 GB), 17 hours of audio, 432 pictures, 18 observation notes, and 15 interview notes. This research applied qualitative methods [17, 24] for data analysis. We utilized iflyre [4] to convert the audio to text, and then corrected the machine-translated text, particularly the part of local dialect spoken by the caregivers. The modified text was then coded using thematic analysis. We first organized the materials from the four studies, noting participation each time a child joined in the game and then coded the key incidents for each participant. Finally, content analysis was used to reclassify the essential events, and the video segments were merged and edited. A total of four researchers conducted the analysis. During the analysis, they read the verbatim transcripts of the interviews, reviewed the video records, summarized the interviewees' attitudes, and described these categories more precisely under themes. Afterwards, they re-examined the overall material and coded it. During coding, the themes were revised and refined to more accurately reflect the information, and these categories were also renamed to reflect the comments of the therapists and the behavior of the participants.

5 FINDINGS

Overall our observations, interviews, researcher video analysis and collaborative video analysis with expert therapists indicated that our MagicBLOCKS system was enjoyed by children and highly welcomed by therapists and caregivers. The findings begin with examining how auditory and visual effects from the augmented reality tabletop interactions were rewarding, motivating and sustained interest. Subsequently, we will describe our themes related to demonstrate how participants interacted as players and as audience members in cooperative and territorial interactions. We report how the MagicBLOCKS system's format facilitated social opportunities and suggested how audience effects affected children who played. These findings suggest that augmented reality tabletop interactions may be used as interventions to assist children with autism in the future in creating opportunities for social interaction.

5.1 Rewards, Motivation and Sustained Focus

The visual dynamics and sound effects in the augmented reality tabletop interaction system were rewarding, as evidenced by children appearing thrilled and showing joy through physical movement such as jumping. For example, P3 jumped throughout several of our interactive games (see Figure 4).

[4]https://www.iflyrec.com/

We discussed this jumping behavior with his mother, who was present during the session. She explained that these excited behaviors helped him release stress and communicate that he was in a happy mood.

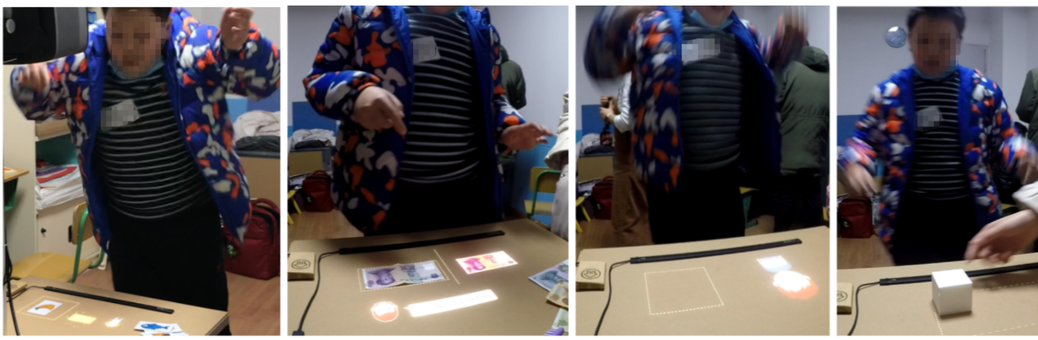

Fig. 4. The animation and sound effects after completing the game mission made P3 excited and he jumped.

The auditory and visual in MagicBLOCKS were seen to be rewarding because they motivated our children participants to practice language skills. Our therapists underlined that language practice could typically be challenging to elicit. During gameplay, the researcher invited children to ask for help and ask questions, which they did in order to trigger the effects. P12 is described as mostly non-speaking because it is difficult for him to verbally express needs in daily life and he rarely speaks in complete sentences. During gameplay, he showed a strong interest in the rotating Ferris wheel dynamics and building blocks. The researcher encouraged him to speak and we witnessed that P12 practised harder than usual, and even used his fingers to make gestures at the same time to express the correct concept (see Figure 5). When the child gives a voice command, the researcher will make an action to make effect occur in the game. The researchers spontaneously customized different challenges for each child to promote their language skills. The result is that in our fourth experiment, many children were spontaneously using language skills. They spoke words related to the physical blocks and cards' like "Doctor", "Firemen", "Teacher", "Panda", "Cat" and "Rabbit". C8 mentioned in the interview:

> "I saw that my son held the building block and practised speaking very hard when he was playing the game. Although it is a few simple phrases/words, I think it is great that he is willing to say it."

Fig. 5. The LFA P12 is obsessed with motion effects and try to speak uses gestures and expressions.

MagicBLOCKS was able to stimulate participants' interest and curiosity in technology. Some low needs participants in this experiment demonstrated this. Take P5 as an example. After completing the game task for the second time, he explored our technology (see Figure 6). He first carefully checked our hardware, projector and laptop, and then asked us about basic technology. Including the device's name, where it was purchased, and what type of device it belongs to; the P5 asked the following questions during the study:

> "Is this a controllable area on the table? Is the light a camera? Is this a button? Did you make it by yourself?"

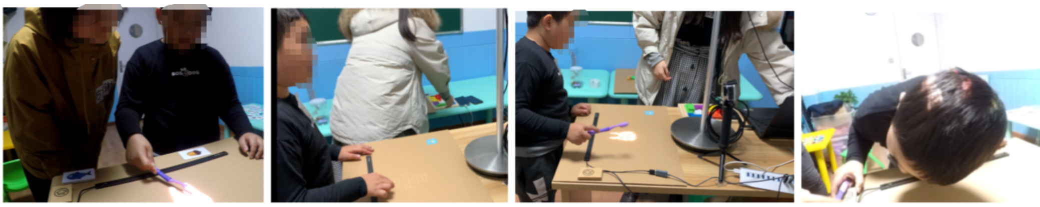

Fig. 6. P5 was interested in the system equipment and asked us about the device architecture.

The motion and sound effects of the projection area were so rewarding that children extended their gameplay. Take P1 as an example. His favourite MagicBLOCKS game was "Puzzle". He played this game repeatedly for more than two-thirds of the session. The therapist explained to us that he wanted to see the repetitive rotation effects in the system. This sustained attention came with learning game specific knowledge. For example (Figure 7), after finishing brushing the Teeth game, he touched his teeth with his fingers. When he sees the firework animation in the system, he also imitates the sound "piu, piu, piu" using his mouth. Th1 mentioned constructive feedback on this aspect. He affirmed the value of the dynamic effects within the system design and said:

> "This kind of strong motion and audio method makes children have a very obvious feeling, which is the kind of very confident and refreshing feeling that I have never seen them show before."

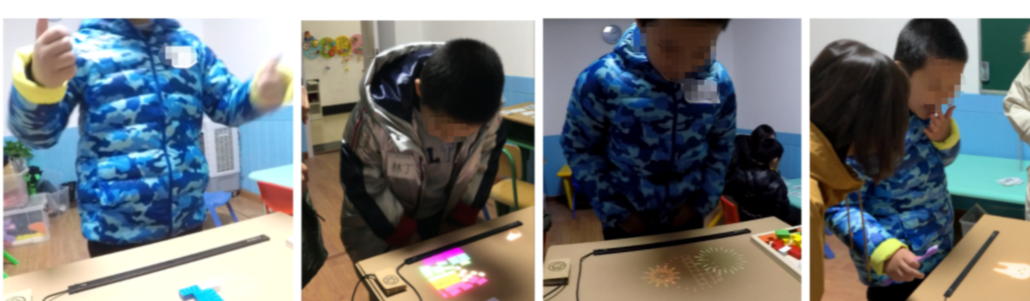

Fig. 7. P1 was interested in the system dynamics and repeated to finish the game task.

We observed that the participants displayed creativity during sustained gameplay with tangible building blocks. Because the placement of physical blocks is based on real-world logic, users have more freedom when placing blocks and are not limited to predetermined play. We set up pattern splicing rules for the children to refer to, but the participants creatively made up their own strategies to complete the game. For example, in the "Puzzle" game, P1 explored three different ways to get the 'completion' experience (see Figure 8).

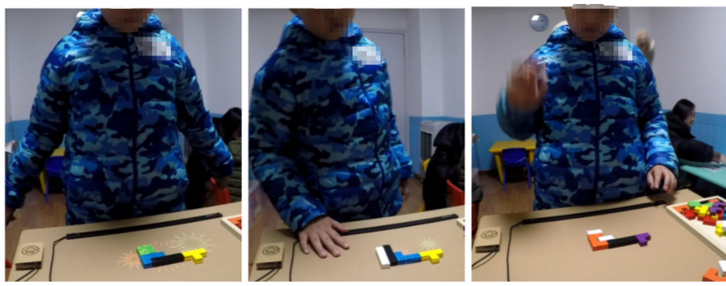

Fig. 8. P1 completing blocks in three creative ways to get the 'completion' experience.

P6 also showed sustained gameplay, repeatedly using tangible objects to complete game tasks (see Figure 7). P6's parents mentioned that this technology is very novel and P6 never tried it before. The therapists interpreted P6's sustained gameplay as a strong desire to use the building blocks to complete tasks to watch the animation effects. Therapists commented that the tabletop format may be beneficial; Th3:

> "The sizes of smartphones and iPad we usually use are relatively small than the table. Many children like to look at electronic screens with closing their eyes and faces close. I think it may damage the children's eyes. Due to the strength of clicking and touching on the electronic screen is almost the same. It's difficult for children to perceive the different touch feedbacks, so they will repeatedly tap the screen."

C2 observed his son's game process and commented on the benefit of the tangible components:

> "My son has an unconscious and uncontrollable behavior in repetitive shaking of his hands. I think this kind of physical interaction game is more suitable for him. I found when he holds a physical object in his hand, those shaking activities were decreased and his learning ability will become more focused."

Fig. 9. P6 was attracted by the motion effects of augmented physical objects.

The therapist Th1 confirmed that the motion and sound effects were rewarding to children with autism, and believed that this method carried distinct advantages compared to other rewards used in daily teaching:

> "When they complete the game task, the animation effect can provide them with strong and novel visual stimulation. I believe that this method of remuneration instills them with more confidence than our traditional verbal or physical encouragement. We also hope to provide this type of reinforcement to children, but it isn't easy to do so in real practice. "

Some parents told therapists that their children showed a strong willingness to use MagicBLOCKS again. As Th2 said:

> "Several parents also reported to me the children's response after participating in the experiment. They mentioned that almost every child would like to participate in the game again. In my opinion, the children feel happy during the whole process of participating in your interactive game."

Overall, we found that the rich auditory and visual effects of augmented reality tabletop interaction were rewarding for children with autism. Next, we turn to how MagicBLOCKS enabled social interaction between children with autism. We found two main types of interactions: collaboration and examples of territoriality. We explain each next.

5.2 Collaboration

One lens for understanding the social interactions facilitated by the MagicBLOCKS tabletop gameplay is to look at the role as a player and the role as an audience member. Audience members included children with autism, family members, operators and on-site volunteers. On average, 1-2 audiences gather around the player. If the player only repeats a game, the audiences may leave. As we had designated turn-taking and which child would have the next turn, sometimes only the next child would watch the entire gameplay of the current player.

Our video data indicated that when children with autism played, their general attitude toward the audience was friendly and tolerant. Player P5 attempted to introduce the current "Occupation" game to the audience, and he says "The flames here are on fire" (Figure 10. A). With his fingers on the physical blocks, audience member P8 stood on the adjacent side of the table and said "firefighter" (Figure 10. B). "Firefighters can put out the flames," P5 continued (Figure 10.C). The two of them are attempting to explain the plot of how firefighters put out fires to nearby audiences. As a result, it appears that this tabletop interactive game can stimulate children with autism to practice verbal communication, which is very welcome by therapists and parents.

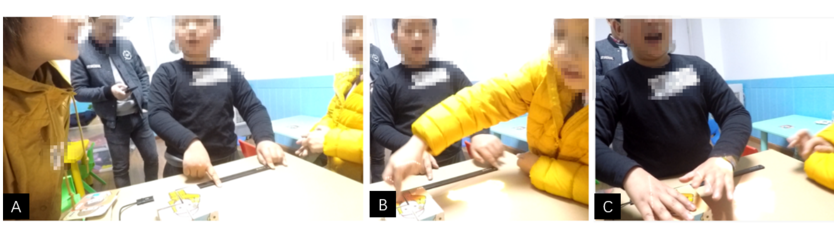

Fig. 10. The player P5 interacted with the audience member P8 throughout the "Occupation" game (A, P5 says there is a fire; B, P8 points to the building block and says that the pattern on it is a fireman; C, P5 says that the firefighter can put out the fire).

We observed that children with autism seemed more inclined to accept suggestions from child audience members. For example, P11 "attacked" the hearts that could be used to earn points when playing the "Monster" game. The researchers reminded him not to attack the red heart, but P11 continued with that behavior. After a child audience member reminded him not to attack the hearts, the player stopped attacking the hearts (Figure 11. A). We observed similar social interactions during other games. For example, in the "Puzzle" game, P1 as the player chose inappropriate blocks and could not complete the task. After a bystander reminded the player that he had chosen the wrong brick, the player adopted his suggestion by choosing the appropriate building block and completed the task (Figure 11. B). In an "Occupation" game, player P9 appeared to randomly put the blocks together instead of splicing them according to the rules. Audience member P11 reminded him to ensure that the patterns on the blocks matched when placing the blocks. Then, P9 rearranged the order of the blocks according to the pattern to complete the task (Figure 11. C).

Fig. 11. Audiences' help the player. A: the audience reminds P11 that he should not attack the red heart, and then P11 stops attacking. B: the audience points out to P1 that he has chosen the wrong building block and then P1 chooses the correct block to complete the task. C: P11 points out to P9 that he did not notice the pattern on the blocks, and then P9 recombines the blocks to complete the game.

MagicBLOCKS provides opportunities for audience members (i.e., peers) to offer help to players, and for players to be receptive to this help and to cooperate with peers to complete tasks. We describe a few examples of constructive assistance. When P5 found that P9 was unable to complete the building block task, he assisted in locating and placing the correct answer prompt card in the visible area of the tabletop game (Figure 12. A). P4 assisted P9 in locating and placing the correct matching card in the recognition area (Figure 12. B). We identified instances of player-audience collaboration: for example, P7 and P1 tapped the table area together to launch cannonballs and completed game tasks (Figure 12. C). P8 and P11 used the menu interface to select games collaboratively (Figure 12.D).

Fig. 12. The player and the audience member complete the task together (A, P5 provides P9 with hint cards to help him complete the task; B, P1 helps P7 to launch a cannonball; C, P4 helps P9 to place cards; D, P8 helps P11 to choose the game).

Augmented reality tabletop interaction like MagicBLOCKS may provide opportunities for children with autism try to engage in collaborative gameplay with their peers, even if it is challenging to communicate verbally. Figure 13 shows a case in which P7, who is mostly non-speaking, first plays the game alone, and then P1 enters the space. P7 glanced at the candy box in his hand, then p7 stepped to the side as an invitation for P1 to join, and also gestured with his eyes that P1 could join in and play the game. P7 and P1 played together, for a short period. We interviewed P1's mother after the gameplay. She was surprised at the collaboration. She explained that P7 is a bit irritable, will attack peers in daily training, and finds it difficult to share with other children. We followed up on this case with Th1 in the post-hoc interview. He believed that it was caused by a common interest, as he said:

> "I saw this video and my biggest feeling is that having a suitable platform is particularly important. Because they have common interests, they have indispensable connections and allow them to have this so-called peer socialization. I found that in our centre that allowing them to maintain common interests is actually relatively few. Sometimes the therapist may be forcing them to communicate through body and eye contact, the result is they can't see their natural initiative."

Fig. 13. P7 actively shares the tabletop game area with P1.

Our cases indicate that when there are audiences watching a player play MagicBLOCKS, children with autism can show a friendly attitude as "players", and they will also respond to the audience's suggestions and help. Th2 described the benefit of games like MagicBLOCKS offerings social intervention opportunities for children with autism:

> "Whether or not friendly or antagonistic behaviors are normal in younger age, it is critical that we can use them to guide appropriate interventions. We have the opportunity to practice these behaviors only when they occur in real social situations. For instance, we can instruct children how to share, cooperate with their peers, or reject."

5.3 Territoriality

The last section described cases where players and audience members collaborated. We also found ways that players rejected interjections, i.e., 'help', from others. We want to emphasize that rejecting interjection is not necessarily problematic. A child rejecting help is making a choice and exerting control, which can be a constructive and important experience. We use the lens of territoriality, an established concept applied to tabletop interactions [36, 46], to discuss cases where audience interjections were rebuffed by the player. We gathered our cases into three main themes to discuss behavior of audiences and players. First, audiences tried to use tangible objects to interfere with the player's gameplay. Second, the audience used their bodies to seize the player's territory. Third, the player tried to push the audience's hand away.

We found several cases where an audience member used physical game props as bids to participate in the gameplay, which interfered with the players. As shown in Figure 14, the audience tried to use cue cards, building blocks, and candy boxes to grab the players' space. In these cases, audience members did not try to grab anything from the players' hands, but they took other physical objects, which led to blocking the player on the table.

Fig. 14. Audiences used physical objects to try to participate in the gameplay; this interfered with players' gameplay.

We found that several child audience members used their bodies to interject into the gameplay process. For example, an audience member used his hand to pull the player's hand, to drag the player's clothes, or to "squeeze away", i.e., push away, the companions with their own body (see Figure 15). When we discussed these cases with our therapists and parents, they explained that these children typically did not express requests verbally. They explained that it is difficult to express social requests in a manner that appears polite, and it is difficult to wait patiently for the player to complete the task or to complete their turn.

Fig. 15. The audience members use their bodies to enter the main player's game area.

When an audience member interrupts the player in the game, the player may accept their interjection or they may resist. We describe the forms of resistance and rejection that we observed and discussed. Some children with autism are mostly non-speaking and thus do not use verbal language to communicate. Instead, they communicate their desire physically, in some cases by pushing away the audience's hands and body with their hands. Figure 16 shows two cases where the player pushed the audience member's hand away, however this was not always effect as the audience member would reintroduce their hands repeatedly. This happened several times on the game choice menu page. Different children had particular specialized interests in which game to play, and players wanted to make their own decisions. In this case, they audience intervention in the choice of game. Therapists and parents suggested that some children's desire to repeat games may be related to predilections for repeated behavior.

Fig. 16. Players use their hands to push away the interference from the audience.

The findings on Collaboration showed examples where audience interjections were welcomed and integrated into gameplay by the player. In this section on Territoriality, we see that players will sometimes opt to reject interjections. We see this as the player acting appropriately within their role and asserting control over the space that is meant for them when in a player role. We saw examples of physical interjections where audience members' bids to participate came in the form of seizing the tangible objects or reaching into the gameplay area. Sometimes the effects of these physical interjections were that the player's actions were hindered or blocked. In our interviews with therapists, we learned more reasons why this kind of conflict-related behavior can be constructive: because the therapist can use these moments to teach children how to deal with conflicts, as described by Th2 constructively:

> "I believe that this emotion is natural regardless of whether their social behavior is friendly or unfriendly during the game. This is important...because social skills must be practised in real life in order for them to try to connect with strangers. However, it is a challenge for our institutions and only-one-child families."

Over the four sessions, four audience participants were temporarily required to leave the study space due to distressed behaviors which interrupted the player: P4 had three breaks, P5 had two, P8 and P1 had once each. We did not see any associations with distress, age or support needs. No participants left the study due to distress.

This section on Territoriality involved mostly non-verbal communication. We noticed that children with higher needs played MagicBLOCKS solo, i.e., without collaboration, more often. This may reflect a preference or may reflect barriers to engaging with others collaboratively. Some participants with higher needs took longer to play and had more periods of deliberation between episodes of active gameplay. We found that the interjections sometimes happened during these moments of quiet deliberation, and the child rejected interjections at this time. At the point of these interjections, the child sometimes expressed the need for more time and the desire that their turn not be cut short. Participants with lower support needs were more likely to accept the 'help' in the form of audience interjections and the lower needs players tended to play more continuously.

6 DISCUSSION

The primary findings in this paper suggest that MagicBLOCKS animation and sound effects are rewarding to children with autism, and that MagicBLOCKS provides opportunities for peers to engage with each other socially. The augmented audio-visual effects around tangible objects in a tabletop setting successfully supported motivation and sustained interest. Children with autism engaged with the tangible games and with each other through player and audience interactions with episodes of cooperation and territoriality. In the following sections, we will discuss augmented tabletop interaction as a reward for children with autism, as well as how to configure games to enable children with autism to 'try on' different roles thus establishing novel opportunities for social interaction.

6.1 Augmented Tabletop Interaction for Children with Autism

Our MagicBLOCKS games were designed with different levels of difficulty, ranging from simple to difficult, to match the various support needs of our participants. Our observations follow Washington et al. [22, 50, 51] that the ability of children with autism to complete more complex tasks was related to the children's support needs. We observed participants tenaciously trying different games to complete the tasks, and enjoying the visual and auditory rewards. These rewards are one of the strongest similarities to typical multimedia solutions like videos and iPad games — stimulating audio-visual effects. MagicBLOCKS differs from digital games because the system was designed based on tangible tabletop game objects which are common in children's daily life (e.g., toy blocks, physical toothbrushes, banknotes, cognitive cards, colored balls, etc.) which were augmented with digital animation and sound effects.

One motivation to explore tabletop interactive objects was to elicit associations with other life experiences. In other words, the design of interactive technology must consider knowledge transfer in addition to entertaining gameplay. This is especially relevant for children with autism because they generalize differently, making it less likely for them to associate new stimuli with past experiences [33]. Our design followed the recommendations of Sonne and Jensen [39] that when designing games for children with neurological disorders, it is necessary to focus on balancing the participation, difficulty, and visual and auditory stimuli of the game to relax without feeling nervous or frustrated. Therapists felt that the task demands of MagicBLOCKS were acceptable, so children were not overstimulated.

Previously, researchers believed that children with autism might have a lack of imagination and creativity [7, 31]. The videos of our study recorded the creativity of children with autism in flexible gameplay, and we had in-depth discussions with expert therapists focusing on this, indicating that they observed a great deal of creativity. Through tenaciously attempting to find new ways to elicit digital rewards, we were glad that MagicBLOCKS was able to draw out creativity from our participants. It may be that creative tasks must be redesigned with the needs of neurodivergent children in mind to elicit creativity better.

6.2 Interaction among Children with Autism

There is limited previous work that has explored the interactive behavior of children with autism with the lenses of player performances and audiences. By performance, we mean how the presence of a particular audience and a certain role may impact how the performer acts [16]. The results of this study filled this gap. Our observational analysis showed that children with autism in the role of an audience member wanted to participate. They spontaneously verbalized instructions and they spontaneously reached in. When these children were the main players, they were in a position to accept or reject this aid. It was fascinating to see the near-instantaneous rejection of aid, as it was clear that participants enjoyed the animations that occurred during successful completion. The main players who refused the help, with a fast block of their arm were given an opportunity to enact this power, in their role as the main player, was a key example of territoriality [36, 46]. The verbal 'help' also gives us a clue, as the main player might ignore the help or may choose to follow it. Children with autism are not always afforded the opportunity of being in control with peer-play, especially when a lower-needs or dominating personality can take control of gameplay.

The combination of tabletop game design enabled children to have the power of the player role as a performer or presenter, which we were very glad to provide. The role of audience members also has special characteristics. The audience role enables the children to gain familiarity with the game. It gives them an opportunity to cheer on and support the main player, or to attempt to provide help. However, the audience role means that the child is not in charge of gameplay and so must exert some self-control. This practice of self-control is crucial and will be a very helpful skill in interacting with neurodivergent as well as neurotypical peers. The therapists in the post-hoc interview pointed out the importance of this close-range social intervention as real training, in particular how to share with their peers and how to correctly handle conflicts in social relationships. Therefore, we suggest that future researchers and educators should design the configurations of gameplay objects for children with autism to enable them to experience different roles and practice "appropriate" social interactions that can shift depending on the context.

Future work should also explore how to support other forms of performer-audience interactions beyond what we currently support, and how to shape pro-social behaviors during such interactions, probably through example-setting or persuasive design. The configurations of the objects matter. Klinkhammer [23] pointed out that the location and direction of tabletop digital notes can coordinate group activities and influence its task execution and results. Our study also found that the collaborative game rules on the tabletop can promote users' social interactions and opportunities for collaboration, but the limitation is that most participants stood on the same side. We would like to pursue this research in larger spaces to be able to observe how children with autism might place themselves when given more choices for location.

6.3 Limitations and Future Work

While our research provided several useful insights for practice social behaviors of children with autism, it still has several limitations. First, the children who voluntarily signed up were all male, and thus we may have observed different social dynamics with a more diverse group. The therapists thought that boys were naturally drawn to digital interactive products. Because our sample is male-skewed, the findings may not be generalized to female children with autism or children who were not interested in digital interactive systems. Additionally, the experimental settings involved research assistants and children's caregivers, which may affect children's natural behaviors in game-playing and socialising. Children with autism can be more sensitive to crowds. We are unable to disentangle the effects of the additional people in the room with the effects of interacting with MagicBLOCKS. The caregivers and therapists all pointed out in the interviews that their children had never tried such tabletop interactive games before this game, so they were novel for all our participants. While all children had some experience with videogames, there may be a novelty effect and participants behavior may change over longer periods of time. Although caregivers and therapists reported that children demonstrated enjoyment, the system's effectiveness in assisting children with learning requires additional quantitative evidence. Future research could explore a wider range of populations and genders and compare the differences between these groups. An important limitation of our study was that we didn't interview children with autism about their experiences. In the future work, it is critical to obtain feedback of children with autism on their experiences of the system. In addition, this study has found some evidence that tabletop interaction technology supports social practice in neurodivergent children. It might be beneficial to collect more quantitative data and conduct controlled experiments to explore the effectiveness of interactive tabletop technology in supporting neurodivergent children.

7 CONCLUSION

Our research found that the MagicBLOCKS augmented tangible tabletop game system provides a powerful incentives and interactive opportunities for children with autism. Our research indicates that MagicBLOCKS provides children with autism opportunities to engage collaboratively with each other, as well as to assert control as the player, i.e., to resist and reject the audience's bids to participate or to take over gameplay. The augmented tabletop game effects seem to support motivation and sustained interest. MagicBLOCKS was able to provide these opportunities for cooperation and territoriality while making tangible gameplay enjoyable. Our study had limitations, as all our children were male and we did not have additional debriefs with the children on their experiences. The findings presented here alongside with the insights can be beneficial to educators and autism specialists who are interested in exploring how tabletop interactive games can be used in autistic center settings. These results may also be helpful to HCI researchers who are interested in creating technologies for this area.

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